

Text2Video: Automatic Video Generation Based on Text Scripts

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ABSTRACT

To make video creation simpler, in this paper we present Text2Video, a novel system to automatically produce videos using only text-editing for novice users. Given an input text script, the director-like system can generate game-related engaging videos which illustrate the given narrative, provide diverse multi-modal content, and follow video editing guidelines. The system involves five modules: (1) A material manager extracts highlights from raw live game videos, and tags each video highlight, image and audio with labels. (2) A natural language processor extracts entities and semantics from the input text scripts. (3) A refined cross-modal retrieval searches for matching candidate shots from the material manager. (4) A text to speech speaker reads the processed text scripts with synthesized human voice. (5) The selected material shots and synthesized speech are assembled artistically through appropriate video editing techniques.

CCS CONCEPTS

• Information systems → Multimedia content creation.

KEYWORDS

text2video, video generation, video editing, video dubbing

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1 SYSTEM ARCHITECTURE

The system diagram of Text2Video is illustrated in Figure 1. The whole system can be divided into five parts: material manager, natural language processing (NLP), cross-modal retrieval, text to speech, and video editing.

1.1 Material Manager

Modalities of material range from text, images, audios, and videos. Material manager is to manage the material for game video production. All material would be tagged with two low-level labels, namely

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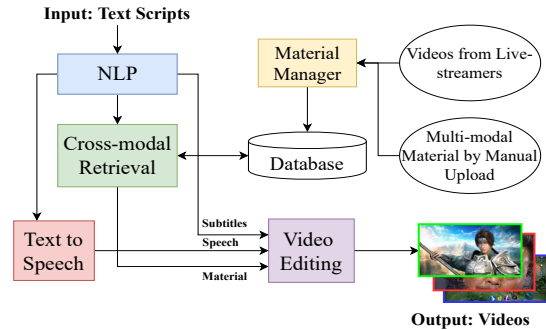


Figure 1: System diagram.



Figure 2: Highlight extractor.

“game-related” and “game-unrelated”, and a series of high-level labels. Labels of each material are provided by human annotation and algorithm classification. The labels and text description of each material will be used in the cross-modal retrieval.

Notably, a large part of the video material are produced by a highlight extractor automatically. Highlights are video episodes where target events of a game happened, such as “first blood”, “double kill”, “triple kill”, “quadra kill” and “penta kill”. Highlights are continuously extracted from raw long videos of game live websites. As shown in Figure 2, the timely broadcasts of game videos provide information about target events. At first, each raw video is split into frame sequences with a frame rate of 2, then we adopt feature points matching [4] and OCR [8] to process each frame and detect the target events. Moreover, each hero has a blood bar overhead, we use template matching [6] to locate the blood bar and a trained CNN models [3] to recognize each hero from 101 heroes with images within ROI (region of interest) under the blood bar. Each highlight with its labels will be stored in our database for cross-modal retrieval.

Title: 爱在三国 (Romantic of Three Kingdoms)

小乔和大乔在郊外玩。遇到了帅气的周瑜。小乔爱上了周瑜，虽然偶尔吵架打闹。直到一个叫做诸葛亮的男人出现。周瑜被他带领的大军打得溃败，郁郁而终。小乔开始追着诸葛亮为夫报仇。

Xiaoqiao and sister Daqiao played in the countryside. Xiaoqiao met the handsome Yu Zhou. Xiaoqiao fell in love with Yu Zhou and flirted with him. Unfortunately, a man named Liang Zhuge appeared. Yu Zhou was defeated by his army and died in depression. Xiaoqiao started chasing Liang Zhuge to avenge the death of her lover.



Figure 3: An example of generated video. The video uses game material to retell a classic historical story.

1.2 NLP

This module is to understand the text scripts. It first splits the input text scripts into sentence sequences, then extracts game entities, namely hero names in our cases, from each sentence. Moreover, each sentence is transformed into a vector embedding by BERT [1]. The entities and vector embedding of each sentence will be used in cross-modal retrieval, and each sentence will be used as a subtitle.

1.3 Cross-Modal Retrieval

Cross-modal retrieval is essential to Text2Video system, which selects the best and closest matching materials, namely video episodes and images, from the database with the entities and embedding from NLP module for each sentence. It first recalls candidate material via Elasticsearch by entities. The video episodes and images which have been tagged by the entities will be selected as candidates. Next, we ranks the candidates by semantic matching through Faiss [2]. More specifically, we compute the embedding similarity between each sentence and the text description of each candidate material. Finally, the video episode and image with the highest similarity for each sentence are selected.

1.4 Text to Speech

Video dubbing for subtitles is indispensable for video production. This module is to generate speech for each sentence from the NLP module. Each piece of speech will be used as a subtitle dubbing. We implement Style Token of Tacotron-2 model to read each sentence [5, 7]. The model is trained on a mandarin speech dataset named THCHS-30. The number of the style tokens is set to 10, and training stops after 55000 steps.

1.5 Video Editing

After the subtitles, dubbing, images and video episodes are prepared, the Text2Video system assembles them according to various predefined video editing templates. A template is a pipeline to organize the multi-modal material sequences with different video editing techniques. For example, transitions would be added to smooth the video switch. Video effects, such as stickers, light and animation, would be used for decoration. We also take advantage of montage

methods including speed adjustment, clip loop and background music implantation. The selection of templates can be determined by users via a configuration file or automatically chosen by machine with a probability distribution. Note that depending on applications, video edition techniques to generate videos can easily be updated.

2 DEMONSTRATION AND SIGNIFICANCE

Figure 3 demonstrates an example of the scenario for automatic text-to-video generation. The origin sentences are divided by underlines with five colors and five numbers, which map to five video episodes surrounded by boxes with the same color and episode number. The videos match the origin text scripts in multi-view such as heroes labels and text semantics, and narrated by a charming female voice. Furthermore, the transition between two video episodes is smoothed with related pictures, and the events and important heroes are surrounded with animation effects. The video precisely expresses all import points of given scripts and embodies artistic values with video montage methods.

Text2Video bridges the gap between text scripts and multi-modal raw material and lowers the difficult for novice users in video creation. Users only need to focus on the quality of scripts like a screenwriter and Text2Video will generate appealing videos to portray the scripts like a director. The system is also able to produce numerous UGC and PGC videos with acceptable quality, which meets the requirement of video feeds.

3 CONCLUSION

In this paper, we present Text2Video, an automatic system for game video generation given only text scripts. After the material for generation are well prepared by a manager, the entire process can be condensed as follows. First, it uses NLP techniques to get labels and semantics from the input text. Next, a multi-step cross-modal retrieval engine searches for the material that best match the text, and a text to speech speaker synthesizes speech to read the text for video dubbing. Finally, the selected material and generated speech are put together sequentially and artistically via video editing. Game enthusiasts can use our system to produce appealing videos by simply writing their stories down.

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